

TECHNICAL CASE STUDY

# SpillTrace

Satellite Oil-Spill Detection, Drift Hindcast/Forecast &  
AIS-Based Vessel Attribution

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**Smart India Hackathon 2026** — Problem Statement SIH26143

National Technical Research Organisation (NTRO)

Leveraging satellite SAR imagery and AIS data correlation to trace maritime oil spills to their source

RANK, NEVER ACCUSE

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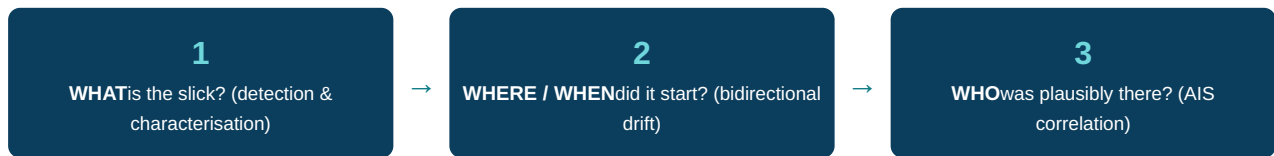
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# 1. Executive Summary

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SpillTrace is a decision-support pipeline that takes one satellite radar image of the ocean and answers three questions in sequence, in the same order a maritime investigator would ask them:



The output at every stage is a **ranked, confidence-scored candidate list with full processing provenance** — never a single accusatory answer. Every number shown on screen can be traced back to the exact code version, parameters, and raw inputs that produced it. This is deliberate: ocean physics and satellite radar do not support certainty, and a system that pretends otherwise produces outputs that cannot be trusted downstream by a real investigator or a court.

## THE STORY, IN ONE PARAGRAPH

A satellite photographs the sea and spots a dark smudge. The system studies that smudge like a forensic scientist studies a stain — checking its darkness, its smoothness, its shape, and its edges — to decide "is this really oil, or just a shadow, a calm patch, or algae?" Once it is confident it is oil, it scatters hundreds of virtual "tracer dots" across the stain and rewinds the ocean currents like rewinding a video, watching where those dots drift back to. That backward cloud of dots — not a single pin — is the most likely birthplace of the spill, in both place and time. Finally, the system checks the shipping-traffic logs (AIS) for every vessel that was near that birthplace at that time, and ranks them by suspicion, showing its work for every single point awarded.

## 2. The Three-Stage Pipeline & Data Sources

| Input                   | Source (this build)                            | Production source         | Status                              |
|-------------------------|------------------------------------------------|---------------------------|-------------------------------------|
| SAR backscatter imagery | Zenodo public oil-spill SAR dataset            | ISRO SAR satellites       | Real mask / synthesised raster      |
| AIS vessel traffic      | NOAA AccessAIS (201 vessels, 24,328 positions) | India's NAIS feed         | Real                                |
| Ocean surface currents  | Synthesised (mean flow + 22 km eddy)           | Copernicus Marine (CMEMS) | Synthesised, structurally realistic |
| 10 m wind field         | Synthesised                                    | ERA5 / INCOIS forcing     | Synthesised                         |
| Ground-truth oil mask   | Official Zenodo test mask                      | N/A — validation only     | Real                                |

The case study region is the Gulf of Mexico off Louisiana (chosen because it has real, dense AIS traffic and a real oil-spill dataset covering it). It is explicitly a **constructed validation scenario**: real SAR imagery and real AIS traffic are co-located onto a common region/time, with one synthetic "ground-truth polluter" vessel injected (AIS-dark for 94 minutes across the release window), so the entire pipeline's accuracy can be measured against a known correct answer. This disclaimer is shown permanently in the product UI, never hidden in a footnote.

### HONESTY CONSTRAINT BAKED INTO THE ARCHITECTURE

Every ML or physics component ships a simple, deterministic fallback *first*. The classical (non-learned) detector is the guaranteed path that cannot fail on demo day; the optional neural network is a strictly additive upgrade layered on top of the same interface, never a replacement for it.

## 3. Stage 1 — Detection: Finding the Oil

### 3.1 Why radar, and why a dark patch means oil

Sentinel-1-class satellites carry **Synthetic Aperture Radar (SAR)**: they beam microwaves down at the ocean and measure how much bounces back. Normal wind-roughened water scatters radar energy well in many directions (called Bragg scattering) and looks bright. A film of oil on the surface physically damps those small centimetre-scale ripples, so far less energy bounces back toward the satellite — the oil-covered water looks like a **dark smudge** against a brighter sea. The raw input to detection is therefore a calibrated backscatter raster in decibels (dB).

### 3.2 The classical detection chain

This is a deterministic, rule-based computer-vision pipeline — no training data, no GPU, cannot fail on demo day:

```
speckle filter (Lee, 7×7)
  → land / hard-target (bright) mask
  → adaptive local threshold (local mean - 0.55σ, 129px window)
  → morphology (open then close, fills gaps/removes speckle)
  → connected components + contours
  → per-region physical discrimination (oil vs. look-alike)
```

The threshold is **adaptive** rather than one fixed global cutoff, because SAR scenes carry a brightness ramp across their width from the incidence-angle dependence of backscatter — a single global cut would either miss real oil on the bright side of the image or flood the dark side with false alarms.

### 3.3 The four physical discriminators (oil vs. look-alike)

Every candidate dark region is scored 0–1 on four physically-motivated tests, then weight-averaged into one confidence score. This is what makes "why oil, why not" auditable rather than a single opaque number.

| Discriminator  | Weight | What it measures                    | Why it separates oil from look-alikes                                                                                                                                          |
|----------------|--------|-------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Contrast       | 34%    | dB darker than local background     | Real oil damps Bragg scattering hard — typically 5–10 dB below the surrounding sea.                                                                                            |
| Variance ratio | 31%    | Speckle suppression vs. ambient sea | Oil kills small-scale roughness, so the patch is not just darker but <i>smoother</i> . A low-wind patch is dark but stays speckled — this is the single most useful separator. |
| Shape          | 23%    | Elongated trail vs. compact blob    | An underway discharge is a long thin trail. Low-wind zones and biogenic (algae) slicks are round blobs.                                                                        |
| Edge sharpness | 12%    | Boundary gradient strength          | An oil boundary is a sharp discontinuity; a wind-roughness gradient fades gradually.                                                                                           |

```
score = 0.34·contrast + 0.31·variance + 0.23·shape + 0.12·edge
is_oil = score ≥ 0.45
```

Regions that fail the threshold are not silently discarded — they are returned as **rejected look-alikes with a stated physical reason** (e.g. "compact form rather than a trail; speckle variance 0.82 of ambient — dark but still rough, characteristic of a low-wind zone, not a damped oil film"). This pre-answers the most common technical challenge to any SAR oil-detection system: biogenic slicks, rain cells, and low-wind zones.

#### SIMPLE EXAMPLE

Imagine looking for a coffee stain on a picnic blanket in a photo. You don't just look for "darker" — you check: is it darker (contrast) AND is the fabric weave flattened (variance) AND is it shaped like a spill-trail rather than a round splash (shape) AND does it have a crisp edge (edge)? A shadow from a tree is darker but the weave still shows clearly through it — so you correctly say "shadow, not stain."

## 4. Post-Detection Characterisation

### 4.1 Geometry

Once a region is confirmed oil, its polygon is converted to real-world units: area (km<sup>2</sup>), perimeter, elongation (major/minor axis ratio), compass orientation of the major axis, and compactness ( $4\pi \cdot \text{Area} / \text{Perimeter}^2$  — 1.0 is a perfect circle). These numbers feed directly into the age estimate below.

### 4.2 Age estimation — how old is this spill?

The problem statement itself hedges spill-age as "if feasible" — there is no accepted method in the literature to recover elapsed time from a single image. SpillTrace attempts it, bounds it as a range, and states its method openly rather than emitting a falsely precise number.

#### THE KEY GEOMETRIC INSIGHT

An underway discharge is a **line source**, not a point source. The trail's *length* is set purely by how far the vessel travelled while leaking — it carries zero age information. Only the trail's *width* carries age, because width grows purely from ocean turbulence spreading the oil sideways over time.

Width is inverted through **Okubo's (1971) empirical scale-dependent horizontal turbulent diffusivity**, derived from real dye-release experiments at sea:

$$\begin{aligned} K &= 0.0103 \cdot L^{1.15} && (\text{cm}^2/\text{s}, L = \text{length scale in cm}) \\ \sigma &= \text{width} / 4 && (\text{visible trail taken as } \pm 2\sigma \text{ across}) \\ t &= \sigma^2 / (2K) && (\text{turbulent spreading time}) \end{aligned}$$

The result is reported as a **range** (e.g. 4.5–19.1 hours), because ocean diffusivity itself varies by roughly a factor of two even at a fixed spatial scale — that uncertainty dominates the error budget and is folded into the bracket rather than hidden. A second, independent proxy — how much the backscatter is damped (dB) — is used only to *shade* the estimate toward the fresher or more weathered end of that bracket, never to set it outright. Confidence is always labelled "low."

### 4.3 Volume estimation

Radar backscatter alone cannot measure oil thickness — thickness is the dominant unknown and there is no accepted inversion from a single grayscale image. Instead of guessing one thickness, SpillTrace uses the same visual-appearance thickness bands real spill-responders use in the field: the **Bonn Agreement / ITOPF Oil Appearance Code** (sheen  $\approx 0.3 \mu\text{m}$  to continuous true colour  $\approx 200 \mu\text{m}$ ), mapping the same contrast-in-dB signal onto a thickness bracket, then  $\text{Volume} = \text{Area} \times \text{Thickness}$ . Reported in m<sup>3</sup>, litres, and barrels, always labelled as an order-of-magnitude estimate, not a calibrated measurement.

## 5. Stage 2 — Hindcast: Finding Origin Point & Time

### 5.1 The algorithm: a stochastic Lagrangian particle ensemble

#### THE STORY

Imagine scattering 500 tiny rubber ducks across the *entire* oil stain — not just one duck at the centre, because a 30 km-long trail obviously did not begin its life as a single dot. Then you play the ocean currents *backward*, like rewinding a video of a river, and watch where every duck drifts back to. That backward cloud of ducks — not a single point — is the most likely birthplace.

Each of the (default 500) particles is advanced in small 15-minute steps using:

```
surface_velocity = ocean_current + wind_factor · 10m_wind      (wind_factor ≈ 2–4%)

step:
  dx = direction · u · dt + √(2·K·dt) · N(0,1)    [advection + turbulent diffusion]
  dy = direction · v · dt + √(2·K·dt) · N(0,1)

  direction = -1 for hindcast (backward in time)
  direction = +1 for forecast (forward in time)
  K = Okubo diffusivity, recomputed every step from the CURRENT spread of the cloud
```

The **wind\_factor** (2–4% of the 10 m wind) is the standard operational range for oil drift — oil floats on the surface, so it is dragged by the ocean current plus a small share of the wind, standing in for the combined effect of Stokes drift and the wind-driven surface layer.

The **diffusion term is the critical modelling choice**: a plain deterministic backtrack would return one exact point and falsely imply a precision the ocean does not support. Adding calibrated randomness — driven by the very same Okubo scale-dependent law used for age estimation, so both stages stay physically consistent with each other — means the spread of the cloud is the answer.

#### WHY REWINDING TIME STILL WIDENS THE CLOUD

Advection is reversed for a hindcast ( $\text{direction} = -1$ ), but diffusion is **not** reversible — turbulent spreading only ever makes a cloud wider, in either time direction. So the further back you rewind, the blurrier (larger) the candidate-origin region becomes. This is not a bug; it is the honest statement that older releases are harder to pin down.

### 5.2 Deriving the origin point and origin time

1. Seed particles across the *whole* detected slick polygon (not its centroid).
2. Advect the ensemble backward for the *maximum* plausible age from Stage 1 (e.g. 19.1 hours).
3. Pool **every particle position** whose elapsed backward time falls anywhere inside the age window (e.g. 4.5–19.1 h) into one combined cloud — this union, not one arbitrarily chosen hour, is the origin region.
4. Convert the pooled cloud into a smooth density surface (Gaussian-blurred 2-D histogram — functionally a kernel density estimate), then find the density contour that encloses 50% and 90% of the total probability mass, traced as GeoJSON polygons. This follows the true curved, sheared shape a real current field produces, which a simple ellipse or convex hull would misrepresent.
5. **Origin point** = the single densest cell of that density surface (the statistical *mode*) — reported only alongside its containment region, never standing alone as a claimed exact spot.
6. **Origin time** = satellite acquisition time – midpoint of the age window.

```
origin_time = detection_time - (age_min + age_max) / 2
origin_point = mode( pooled_particle_density )
uncertainty = { 50% containment polygon, 90% containment polygon }
```

#### SIMPLE EXAMPLE

If you find spilled juice on a floor and don't know exactly when it happened, you don't guess one exact drop point — you draw a "probably here" circle and a bigger "almost certainly here" circle, because the puddle could have been spreading for anywhere from 10 minutes to an hour. That is exactly what the 50% and 90% cones represent for the spill's birthplace.

On the frozen case study, this method recovered the true origin within **7.7 km**, and the 90% containment region did contain the real, known origin point.

## 6. Stage 2 — Forecast: Predicting Future Spread

The forecast reuses the **exact same Lagrangian engine** described in Section 5 — same current+wind advection, same Okubo-scaled turbulent diffusion — with only the sign flipped (`direction = +1`) and the particles re-seeded across the *current* slick shape rather than a historical one. There is no separate forecast model; using one physics engine for both directions is what keeps the hindcast and forecast internally consistent.

| Output                              | What it is                                                                                                                                                                                                             |
|-------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 90% containment cone, per time step | Where the oil is likely to be at each future hour, for animation.                                                                                                                                                      |
| Centroid path                       | A single line tracking the "average" drift of the whole cloud over time.                                                                                                                                               |
| Impact flags                        | If the predicted drift approaches the coastline or a protected area, an estimated time-to-arrival and distance is computed from the cloud's velocity component toward that boundary — an early warning for responders. |

Because diffusion only ever widens the cloud, a forecast necessarily gets fuzzier the further out it looks — exactly the same honest behaviour ordinary weather forecasts exhibit.



## 7. Ocean Physics Reference & Enhancement Roadmap

### 7.1 Forcing field

Surface currents and 10 m wind are bilinearly interpolated from a gridded field (currently a coarse 32×32 synthesised grid; production-ready to accept ERA5/CMEMS NetCDF unchanged). The field is treated as **steady** (time-invariant) over the length of one run — a stated simplification, reasonable because the mesoscale field varies slowly over a 24-hour window in the study region, but explicitly named as the dominant source of error versus using real time-varying forcing.

### 7.2 Core physics equations used throughout the pipeline

```
Surface drift velocity:      v_surface = v_current + wind_factor · v_wind10m
Okubo (1971) diffusivity:   K = 0.0103 · L1.15 [cm2/s, L in cm – scale-dependent: bigger
                             patches are pulled apart faster by progressively bigger eddies]
Random walk step:          Δposition = v_surface·dt + √(2·K·dt)·N(0,1)
Trail-width age inversion:   t = σ2 / (2K),   σ = trail_width / 4
Bonn Agreement thickness:   thickness(contrast_dB) interpolated log-linearly, 0.3–200 μm
```

Using the *same* Okubo diffusivity law for both spill-ageing (Section 4.2) and drift-ensemble spreading (Section 5.1) is a deliberate consistency choice — the age estimate and the drift uncertainty rest on one shared physical model rather than two independently-tuned ones.

### 7.3 How to enhance the physics further

- **Time-varying forcing** — replace the steady field with real ERA5 (wind) and CMEMS (currents) NetCDF time series; the interpolation layer already accepts this without redesign.
- **Explicit Stokes drift & weathering** — separate the folded-in wind factor into a proper Stokes-drift term plus an evaporation/emulsification weathering model over time.
- **Data assimilation** — correct the ensemble mid-flight as new SAR passes of the same spill arrive, rather than running one static estimate.
- **Full physics engine** — swap the ~100-line in-house ensemble for **OpenDrift** once CMEMS credentials and a conda-scale install are available; the codebase already isolates this behind a clean engine seam.
- **Higher-resolution nested currents** near the coastline, where the coarse open-ocean field is least accurate.

## 8. Machine Learning Models Used

SpillTrace deliberately keeps ML usage minimal and strictly additive. Every physics or scoring component ships a deterministic fallback first; the one learned model in the entire pipeline is an optional upgrade to Stage 1 detection.

| Model                         | Architecture                                      | Training data                                                                      | Purpose                                                                                                                                                                    |
|-------------------------------|---------------------------------------------------|------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| U-Net segmentation (optional) | ResNet34 encoder, ImageNet-pretrained, fine-tuned | Public Kaggle "Deep-SAR oil-spill segmentation" dataset (ALOS PALSAR / Sentinel-1) | Learn to paint the oil mask pixel-by-pixel directly from the image, without hand-tuned brightness thresholds — a potential upgrade over the classical rule-based detector. |

### WHY THE NETWORK IS GATED, NOT TRUSTED ALONE

The checkpoint was trained on a *different* SAR collection than this case's calibrated raster, so a percentile stretch bridges the value ranges — a documented heuristic, not a validated equivalence. The training data also has no separate "look-alike" class: on its own, the network confidently flags round calm-water and biogenic patches as oil (measured: 100% recall but only 5.9% precision before gating). The fix is a **shape plausibility gate** reused from the classical detector — a region is only kept as oil if it is trail-shaped (low compactness), not round — because the classical detector's full four-term score doesn't transfer to a U-Net-shaped boundary (its variance term measured the true slick as *rougher* than background on this scene, the physical opposite of oil damping — a boundary-measurement artefact, not a real signal). Regions the network liked but the shape gate rejects are still surfaced as rejected look-alikes, with the override reason stated.

Every other stage of the pipeline — geometry, age, volume, drift, and attribution scoring — is deterministic physics or explainable weighted arithmetic, not a learned model. This is intentional: it keeps every number on screen traceable to a stated formula rather than an opaque network weight.

## 9. Stage 3 — Attribution: Relating Origin to AIS Vessel Data

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### 9.1 What AIS is

**AIS (Automatic Identification System)** is a mandatory maritime beacon: every large ship broadcasts its GPS position, speed, and heading every few seconds to minutes. This build uses NOAA's AccessAIS feed (201 vessels, 24,328 positions in the case region); production would use **India's NAIS feed**.

### 9.2 Technical approach

1. Take the **origin point + origin time** produced by the hindcast (Section 5).
2. Define a space-time search box around it: default **25 km radius, ±6 hour window** — both are tunable parameters in the API contract.
3. Pull every vessel track that intersects that box, and reconstruct each ship's path from its raw AIS position reports.
4. **Gap detection**: flag any silence in a vessel's AIS transmissions longer than a threshold (30 minutes) — a "dark" period during which the ship could have been discharging while deliberately hiding its position.
5. Score every candidate vessel on five independent, transparent 0–1 factors (Section 10).
6. Rank the candidates and generate a plain-English narrative sentence for each, built directly from its score breakdown, so nothing on screen is an unexplained number.

#### SIMPLE EXAMPLE

It's like a teacher figuring out who spilled juice in a hallway. She doesn't blame the nearest student — she checks who was standing nearby (proximity), who was there at the right time (temporal overlap), whose walking direction matches the juice trail (heading), who suspiciously vanished right when it happened (the AIS gap), and who was walking oddly slowly (speed anomaly). Then she ranks her suspects and explains her reasoning for each, rather than punishing anyone off a single clue.

## 10. Confidence Scoring Methodology

Every vessel candidate receives five independently-computed 0–1 sub-scores, combined by a transparent weighted sum — never a black-box single number:

| Factor              | Default weight | What it captures                                                                                                                                                    |
|---------------------|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Proximity           | 30%            | How close the vessel got to the origin point. Scores 1.0 at 0 km, decaying linearly to 0 at the search radius (25 km).                                              |
| Temporal overlap    | 25%            | Whether the vessel was actually present <i>at the estimated origin time</i> , not merely nearby at some unrelated hour.                                             |
| AIS gap             | 20%            | Whether the vessel's beacon went dark near the origin window — a possible sign of deliberately hiding a discharge. Longer, better-placed gaps score higher.         |
| Heading consistency | 15%            | Whether the vessel's compass heading around that time matches the drift/trail direction of the slick — a ship steaming north cannot explain an east-pointing trail. |
| Speed anomaly       | 10%            | Whether the vessel was moving unusually slowly for a while — consistent with "slow-steaming while discharging" rather than normal transit.                          |

```
confidence_score = 0.30·proximity + 0.25·temporal_overlap + 0.20·ais_gap
                  + 0.15·heading_consistency + 0.10·speed_anomaly

rank = sort candidates by confidence_score, descending
```

All weights are configurable per-request in the API contract, and every response returns both the final score *and* its five ingredients individually, so a reviewer can see exactly why a vessel ranked where it did — for example: *"passed within 6.4 km of the origin; heading closely matches the drift direction; 38-minute AIS gap detected near the spill window."*

**DESIGN RULE: RANK, NEVER ACCUSE**

The output is explicitly labelled a ranked, confidence-scored *candidate list*, never an identification. AIS can be switched off or spoofed for many innocent reasons (poor signal, equipment fault), so a gap is treated as **one weighted suspicion signal among five, never as proof**. The physics and the traffic data simply do not support certainty, and the system is designed to say so on screen rather than in a footnote.

# 11. Measured Results

Measured against the frozen gom-2023-06-15 case study, scored against the official Zenodo ground-truth mask:

0.878

DETECTION IOU

93.1% / 93.9%

RECALL / PRECISION

7.7 km

HINDCAST ORIGIN ERROR

38 ms

DRIFT ENSEMBLE RUNTIME

| Metric                         | Value                                      |
|--------------------------------|--------------------------------------------|
| Detection IoU                  | 0.878                                      |
| Recall / precision             | 0.931 / 0.939                              |
| Detected area vs. truth        | 15.9 km² (truth 16.0 km²)                  |
| Orientation vs. truth          | 65° (truth 65°)                            |
| Look-alikes correctly rejected | 2 of 2, each with a stated physical reason |
| Age bracket vs. truth          | 4.5–19.1 h (truth 8.0 h)                   |
| Hindcast origin error          | 7.7 km (mode vs. true origin)              |
| 90% origin containment region  | Contains the true origin                   |
| Drift ensemble runtime         | 38 ms                                      |

The classical detector has no learned weights and is fully deterministic — these numbers reproduce exactly on every run of `pytest tests/test_detection.py`.

## 12. Design Principles & Current Build Status

### 12.1 Load-bearing design rules

1. **The demo runs offline.**

No live API calls, no CDN, no network basemap — conference wifi has killed more demos than bad code.

2. **Rank, never accuse.**

Every output is a confidence-scored candidate list with a per-factor breakdown; the physics does not support certainty.

3. **Fallback before headline.**

Every ML or physics component ships a simple deterministic version first. The classical detector is the guaranteed path; the U-Net is a cuttable upgrade.

4. **Caveats go on screen, not in a footnote.**

The age estimate carries its method note, the AIS gap carries its base-rate warning, and the constructed-scenario disclaimer is a permanent, non-dismissible banner.

5. **Uncertainty is a cone, not a pin.**

The hindcast returns 50% and 90% containment polygons; the origin marker is only the mode of the ensemble.

### 12.2 Build status

| Phase | What                                                     | Status                                                                     |
|-------|----------------------------------------------------------|----------------------------------------------------------------------------|
| 0     | Scaffold, frozen API contract, fixtures, dashboard shell | Done                                                                       |
| 1     | Freeze the case-study data bundle                        | Done                                                                       |
| 2     | Detection, geometry, look-alike rejection, age proxy     | Done — IoU 0.878                                                           |
| 3     | Drift engine, hindcast cone, forecast                    | Done — origin error 7.7 km                                                 |
| 4     | AIS ingest, gap detection, explainable scoring           | Not started — currently a placeholder scoring engine on mocked vessel data |
| 5     | Dashboard against real data                              | Shell done, fixture-backed                                                 |
| 6     | Evidence PDF                                             | Content endpoint done, PDF pending                                         |
| 7     | U-Net upgrade (optional)                                 | Not started                                                                |
| 8     | Demo hardening and deck                                  | Not started                                                                |

#### CURRENT STATE HONESTLY NOTED

The attribution scoring engine described in Section 10 is the designed, intended approach captured in the project's frozen API contract. As of this document, the actual implementation file runs on a small hardcoded list of example vessels rather than live AIS ingestion, and needs reconciling with the contract's official five-factor field names before it is production-real. Sections 3–7 (detection, geometry, age, hindcast, forecast) are fully implemented, tested, and match the measured results in Section 11.

Prepared as a technical case study for SpillTrace — Smart India Hackathon PS SIH26143. The system is deliberately never an identification: output is a ranked, confidence-scored candidate list with full processing provenance, intended to support — not replace — human investigative judgement.